Volume Based Trading Strategy

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***Abstract*—This paper presents a volume-based trading strategy integrating machine learning techniques, statistical modeling, and rule-based decision systems to predict trading volumes and generate actionable signals. Recognizing the established relationship between trading volume and stock returns, we utilize a Long Short-Term Memory (LSTM) model to forecast trading volumes using historical price and volume data. Signal generation is achieved through quantile-based volume thresholds, market regime classification, and return expectations, enabling adaptive responses to both bull and bear markets. The strategy incorporates dynamic position sizing and risk management mechanisms, including stop-loss and take-profit thresholds, to optimize returns. Backtesting on historical stock data validated the strategy, yielding a Sharpe Ratio 1.34, a net profit/loss of 1,310%, and an effective correlation between predicted and actual trading volumes. Statistical tests, including equity curve test, correlation test, t-test and rolling based sharp ratio test, further support the robustness and significance of the results. This study contributes to bridging the gap between theoretical insights and practical implementation, demonstrating the potential of volume-based trading strategies in modern financial markets.**

I. Introduction

The goal of this paper is to design and present a volume-based trading strategy that leverages insights from trading volume dynamics to predict stock returns and optimize trading decisions. Trading volume, a significant indicator of market activity, has been deeply studied for its relationship with price movements, market liquidity, and investor behavior. Despite these studies, the practical application of volume-based strategies in real-world trading systems is insufficient, especially in distinguishing market regimes (bull and bear markets) and understanding nonlinear relationships between volume and returns. This paper aims to:

- Forecast trading volumes using machine learning models to capture historical patterns and future trends.
- Identify market regimes through dynamic indicators, distinguishing bullish and bearish activity periods.
- Establish a framework for generating signals based on predicted trading volumes, market conditions, and price returns.
- Implement dynamic risk management strategies such as stop-loss thresholds and adaptive position sizing to ensure sustainable performance.
- Validate the strategy through rigorous backtesting on historical stock data and evaluate performance using financial metrics such as Sharpe Ratio, and Net Profit/Loss.

This study aims to create a strong trading framework by integrating data-driven forecasting models and rule-based trading execution. This approach allows it to adjust to changing market conditions while ensuring consistent, risk-adjusted returns.

II. Literature Review

2.1 Relationship Between Volume and Returns

For decades, the bond between trading volume and stock return has been one of the main topics in financial research. Understanding this link is crucial because trading volume is a proxy for market sentiment, liquidity, and the intensity of information flow among market participants.

Theoretical Foundations Early research, including those by Karpoff (1987), offers that trading volume is related to price movements, reflecting investors' responses to recent information and changing market sentiment. Market microstructure approaches suggest that high trading volume usually points more unequal access to information among investors. It also implies higher investor activity, which can affect stock price changes and volatility.

Empirical Evidence

- **Chen (2012):** Shows an asymmetric relationship between stock returns and trading volume across bull and bear markets. In bull markets, returns show a positive correlation with trading volume, because of investor overconfidence and momentum trading. However, in bear markets, panic selling, and risk aversion shifts relationships negatively.
- **Chordia et al. (2001):** Point out that stocks with more unpredictable trading volumes usually have lower expected returns. This goes against the common belief that higher trading activity should cause higher returns because of increased liquidity.
- **Chordia & Swaminathan (2000):** Emphasize that trading volume affects how quickly stock prices respond to market news. High trading volume stocks adjust to new information quickly unlike lower ones. This creates patterns in returns that can be predicted over short periods.
- **Gebka & Wohar (2013):** Utilized quantile regression to study the relationship between trading volume and stock returns. They found that when stock prices rise sharply, higher trading volume tends to push returns even higher.

However, during sharp price drops, higher volume tends to make returns fall further. This shows that the nonlinear relationship between volume and returns is more complex than linear models offer.

Trading volume plays two important roles: it serves as an indicator of market sentiment and acts as a key factor influencing price movements. The relationship between volume and returns can vary across different market regimes in both direction and strength. Understanding these variations requires consideration of nonlinear dynamics and cross-autocorrelations. These observations help build a strategy based on trading volume. This strategy considers market asymmetry, changes in liquidity, and nonlinear causality effects.

2.2 Trading Volume and Expected Returns

Detailed investigations have been made on the relationship between trading volume and expected return. It has been revealed that trading volume plays a crucial role in shaping stock return behavior. Trading volume acts as an indicator of liquidity, information flow, and investor sentiment, each contributing to return dynamics.

Empirical Evidence on Trading Volume and Expected Returns

- **Chordia et al. (2001):** Show that there is a negative relationship between expected stock returns and the variability of trading activity. Their study reveals that stocks with higher variability in trading activity show lower expected returns, even after controlling for size, book-to-market ratio, and momentum. However, normally the initial expectation that increased variability in liquidity would command higher expected returns because of liquidity risk. Moreover, observations from Chordia et al. (2001) emphasize that trading activity metrics, such as dollar trading volume and turnover, are reliable indicators of liquidity. Greater uncertainty and risk for investors can be implied by high variability in these metrics. However, instead of being rewarded with higher expected returns, these kinds of stocks tend to have a return discount.
- **Chen et al. (2012):** Investigates the causal relationship between stock returns and trading volume, emphasizing an asymmetry between bull and bear markets. In bull markets, trading volume is more closely associated with positive price movements. In contrast, during bear markets, trading volume tends to amplify negative price trends.
- **Gebka & Wohar (2013):** Provide additional evidence of nonlinear effects in the volume-return relationship through quantile regression analysis. They demonstrate that trading volume influences expected returns differently across various return quantiles. Specifically:
  - At higher return quantiles, trading volume has a positive impact on expected returns.
  - At lower return quantiles, trading volume has a negative impact on expected returns.

Due to these nonlinearities, traditional linear models cannot fully explain the relationship between trading volume and expected returns. Variability in trading activity is not only the absolute level of volume but also important in determining expected returns. The impact of trading volume on expected returns changes across different market conditions. Nonlinear effects determined by quantile regressions show that trading volume's impact on expected returns depends on the magnitude and direction of price changes. Understanding these relationships is essential for developing a volume-based trading strategy that considers variability, asymmetry, and nonlinear dynamics in trading activity.

2.3 Volume and Market Information Efficiency

The trading volume shows investor activity and is a key indicator of how effectively market information is incorporated into stock prices. This can be useful for developing trading strategies that benefit from how stock prices react to new information.

Empirical Evidence on Volume and Market Information Efficiency

- **Chordia & Swaminathan (2000),** investigate the role of trading volume in the speed of price adjustment to market-wide information. Their findings demonstrate that:
  - Stocks with high trading volume adjust more quickly to market-wide information.
  - In contrast, stocks with low trading volume respond more slowly to the same information, creating predictable lead-lag patterns.
- This lag in absorbing information is not entirely attributed to nonsynchronous trading or portfolio autocorrelations. Instead, it emphasizes a fundamental difference in how information flows across stocks with varying liquidity and trading activity levels.
- **Chordia et al. (2001),** reinforce this observation by showing that liquidity variability impacts the efficiency of market responses. Stocks with greater variability in trading activity show less predictable responses to new information, which can distort price discovery processes.
- **Gebka & Wohar (2013),** add complexity by highlighting nonlinear relationships between volume and returns across different return quantiles. They show that trading volume's impact on market information efficiency is quantile-dependent:
  - In high-return quantiles, volume leads to faster and more efficient price adjustments.
  - Volume may lead to price overreactions or delayed corrections in low return quantiles.
- **Chen (2012),** highlights the differences in market information efficiency between bull and bear markets. In bull markets, high trading volume is usually linked to momentum-driven price efficiency, while in bear markets, increased volume can reflect panic-driven overreactions.

Trading volume has a significant impact on the speed and accuracy of price adjustments in financial markets. High-volume stocks adjust faster to market-wide information, while low-volume stocks show lagged responses. Market efficiency is asymmetric across market conditions (bull vs. bear markets), with trading volume playing different roles in each phase. Nonlinear effects show that trading volume's impact on price efficiency depends on the direction and magnitude of price movements. These findings highlight the importance of integrating trading volume dynamics into volume-based trading strategies, as they offer actionable signals regarding market efficiency and the rapidity of information spread.

2.4 Causality Between Volume and Returns

Understanding whether trading volume causes stock returns or vice versa is crucial for developing an effective volume-based trading strategy. Causality shows whether past values of one variable can provide statistically significant information for predicting another variable. Researchers have used econometric methods such as Granger causality tests and quantile regressions to investigate this relationship.

Empirical Evidence on Volume-Return Causality

- **Chen (2012),** investigates the dynamic causal relationship between stock returns and trading volume across bull and bear markets. The results suggest:
  - Stock returns predict trading volume in both bull and bear markets.
  - However, the evidence for trading volume predicting returns is weaker, suggesting that returns are generally the dominant variable in this relationship.
  - Additionally, the causal relationship is asymmetric. It behaves differently in bull and bear markets, reflecting shifts in investor sentiment and trading behavior during different market regimes.
- **Chordia & Swaminathan (2000),** explore lead-lag patterns and identify a causal structure where
  - High-volume portfolios lead low-volume portfolios, suggesting that high trading activity helps incorporate market-wide information into prices more quickly.
  - This causality is not fully explained by nonsynchronous trading or autocorrelations, emphasizing the essential role of trading volume in market dynamics.
- **Gebka & Wohar (2013),** use quantile regressions to examine the nonlinear causality between trading volume and returns. Their findings reveal:
  - Positive causality in high quantiles: Trading volume tends to predict upward price movements during periods of strong positive returns.
  - Negative causality in low quantiles: Trading volume tends to predict downward price movements during periods of steep price declines.
  - The relationship is non-persistent, meaning it does not cluster over time, limiting its use in consistent forecasting.
- **Gebka & Wohar (2013),** further demonstrate that traditional linear Granger causality tests fail to capture these nuances. They emphasize the necessity of nonlinear techniques like quantile regressions to reveal the complexities of the volume-return relationship.

The causal relationship between volume and returns is asymmetric and varies across bull and bear markets. Stock returns have stronger predictive power over trading volume than vice versa. High trading volume leads to market adjustments, while low trading volume lags behind in processing information. The causality between trading volume and returns is nonlinear, with different effects observed across various return quantiles. Quantile regressions are valuable tools for accurately identifying causal patterns in volume-return relationships. These findings highlight the complexity of volume-return dynamics and emphasize the need for advanced statistical techniques in designing and evaluating volume-based trading strategies.

III. Methodology

3.1 Data Sources and Preprocessing

3.1.1 Data Source: The dataset used in this study consists of historical stock data for Nvidia (NVDA), a publicly traded technology company. The data includes daily records spanning multiple years, sourced from reliable financial market databases. The dataset contains the columns which are Date, Open Price, High Price, Low Price, Close Price,Adjusted Close Price for accounting dividends and also Volume as a final column.

These features serve as the essential input variables for volume forecasting and the subsequent signal generation within our trading strategy.

The data underwent preprocessing and cleaning to ensure it was suitable for statistical modeling and strategy design. This preprocessing involved normalization, segmentation, and feature engineering steps, which will be explained below.

3.1.2 Data Preprocessing: Before any modeling or strategy development, the dataset was processed through several steps to ensure its statistical robustness and consistency.

Data Cleaning: Any duplicate rows should be removed, missing values should be handled by exclusion, and column data formats should be ensured to be correct. **Feature Engineering:** Calculate daily returns using the formula:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where R_t is the daily return and P_t represents the adjusted closing price at time t. Volatility is calculated as a rolling 10-day standard deviation of returns to capture short-term price risk. Volume is defined as the primary target variable for forecasting. Key features such as Volume, Returns, and Volatility have been selected as inputs for the model.

Data Segmentation: The dataset was split into training (80%) and testing (20%) subsets while preserving the temporal order to avoid data leakage. Rolling windows of 10 days were created to prepare input sequences for the LSTM model. Additionally, a 5-day horizon was established for predicting future trading volumes.

3.1.3 Exploratory Data Analysis (EDA): To better understand the dataset's structure and behavior:

- Trading Volume Over Time: Volume patterns to determine anomalies or cyclical trends has been analyzed.
- Return Distributions: Daily return distributions to assess volatility and outlier behavior has been analyzed.
- Correlation Analysis: Relationships between trading volume, returns, and volatility has been explored.

This analysis provided insights into the dependencies between volume and return, which served as the foundation for the development of subsequent models and strategy design.

3.1.4 Relevance to Study Objective: The preprocessing phase created a strong foundation for developing a volume-based trading strategy that aligned with the study's objectives. Specifically:clean, stationary, and appropriately segmented, we can:

- Market Regime Identification: Prepared inputs to identify bull and bear market regimes.
- Causality Testing: Ensured the dataset was ready for causal analysis between volume and returns.
- Trading Strategy Formulation: Delivered clean, normalized, and temporally ordered data for model training and backtesting.

These preprocessing steps allow our strategy to effectively use trading volume dynamics for forecasting volumes, identifying actionable signals, and optimizing risk-adjusted returns.

3.2 Model Design

3.2.1 Overview of Modeling Approach: The main purpose of the modeling phase is to predict trading volumes and to use these predictions as a basis for generating actionable trading signals. This approach combines machine learning techniques and rule-based signal generation, which ensure adaptability under different market conditions.

The modeling process consists of the following key steps:

- **Volume Forecasting (LSTM Model):** Forecast future trading volumes using a Long Short-Term Memory (LSTM) neural network which is a specialized architecture for time series prediction.
- **Market Regime Identification:** Market Regime Identification: Classify market conditions into bull and bear phases using moving average crossovers (SMA_20 and SMA_50).
- **Signal Generation Rules:** Establish rules for buy, sell, and hold signals based on predicted volumes, market regimes, and recent returns.
- **Backtesting Framework:** Evaluate the strategy's performance on unseen test data, ensuring robust validation.

3.2.2 Volume Forecasting Model (LSTM Neural Network): With LSTM model we aim to forecast future trading volume using historical patterns in volume, returns and volatility.

Data Preprocessing for LSTM (LSTM Model): In this step, the data was split into training (80%) and testing(20%) parts. Window size of 10 days was used in orer to create sequential input data from LSTM model. A forecast horizon of 5 days was used to forecast future volumes and the data was normalized using StandardScaler to make sure input and target features scaled consistently.

LSTM Architecture:

- Input Layer: Sequences of 10 day historical data points for Volume, Returns and Volatility.
- LSTM Layer: A single LSTM layer with 16 units and tanh activation function was used to capture temporal dependencies.
- Output Layer: A dense layer producing a single volume prediction for the target horizon.
- Loss Function: Mean Squared Error (MSE)
- Optimizer: Adam

Training Details: Epochs set to 10, Batch Size set to 16.

Model Evaluation:

After training, the model forecasts trading volumes for the test set, generating a series of forecasted volumes (Volume_pred) aligned with the test dates. These predicted volumes are essential inputs for signal generation in the subsequent trading strategy.

3.2.3 Market Regime Identification: Market conditions were classified into two primary regimes which are Bull and Bear Markets.

- Bull Market:Favorable upward price momentum.
- Bear Market: Downward price momentum or market stagnation.

Methodology: A short-term moving average (SMA_20) and a long-term moving average (SMA_50) were calculated on the Close price.

Regime Classsification:

$$BullMarket(1): \quad \mathrm{SMA}_{20} > \mathrm{SMA}_{50}$$

$$BearMarket(0): \quad \mathrm{SMA}_{20} \leq \mathrm{SMA}_{50}$$

This classification offers context for interpreting signals, enabling the trading strategy to adjust to current market conditions.

3.2.4 Signal Generation Rules: **Objective** Generate actionable trading signals Buy, Sell, Hold based on:

- **Predicted Volume** (Volume_pred)
- **Market Regime** (Regime)
- **Recent Returns** (Returns)

Volume Thresholds:

- **High Volume Threshold:** 90th percentile of predicted volumes.
- **Low Volume Threshold:** 10th percentile of predicted volumes.

Signal Logic

Buy Signal (1):

- Volume_pred $>$ High Volume Threshold
- Market is in a **Bull Regime**.
- Recent Returns > 0

Sell Signal (-1):

- Volume_pred $>$ High Volume Threshold
- Market is in a **Bear Regime**.
- Recent Returns < 0

Hold Signal (0):

- All other conditions.

These rules align volume predictions with broader market behavior, ensuring a strategy that is both responsive to market shifts and rooted in statistical insights.

3.2.5 Backtesting Framework: Evaluate the trading strategy using historical test data and measure its effectiveness with performance metrics. **Key Components:**

- Initial capital is 10,000$
- Position Management: Buy, sell and hold positions based on generated signals.
- Stop-Loss Mechanism: Losses are limited during adverse movements.
- Take-Profit Mechanism: Realize gains upon favorable price changes.

Performance Metrics: Our performance metrics are Sharp Ratio to measure risk-adjusted returns and Net P&L to measure profit/loss across test period.

The backtesting framework assesses the robustness of a strategy and ensures its predictive consistency by using only test data.

3.2.6 Model Validation: The validation phase is focused on evaluating the model's predictive performance and robustness.

- Evaluate the model's performance solely on previously unseen test data.
- Robustness Checks: Evaluate the consistency of results across different parameter settings and market regimes.

3.2.7 Relevance to Study Objective: Each model serves a distinct purpose in addressing the objectives of this project:

- Predict Future Trading Volumes: Use an LSTM neural network to forecast volumes accurately.
- Market Regime Awareness: Classify market regimes into bull and bear phases for adaptive decision-making.
- Signal Reliability: Generate buy, sell, or hold signals based on statistically validated thresholds and the market context.

This detailed model design ensures the strategy effectively captures volume dynamics and translates them into actionable trading insights.

3.3 Strategy Design

3.3.1 Objective of the Strategy: The main goal of this strategy is to exploit trading volume dynamics to make informed trading decisions and optimize risk-adjusted returns. This strategy combines LSTM-based volume predictions, market regime identification, and rule-based trading signals to create a detailed approach to trading NVDA stock. The strategy aims to:

- Identify actionable buy, sell, and hold signals based on trading volume thresholds, predicted volumes, market regimes, and return expectations.
- Adapt to market conditions by distinguishing between bull and bear markets, allowing tailored responses in each scenario.
- Optimize risk-adjusted returns by incorporating dynamic position sizing and stop-loss mechanisms.

This approach utilizes machine learning models, econometric tools, and statistical insights to ensure robustness, adaptability, and alignment with market conditions.

3.3.2 Strategy Framework: **Signal Generation:** The trading strategy that generates signals using insights derived from LSTM volume predictions, moving average regimes, and recent return trends.

Volume-Based Signals:

- **High Volume Surge:** If predicted trading volume (Volume_pred) exceeds the 90th percentile, it indicates significant market activity, warranting attention.
- **Low Volume Drop:** If predicted trading volume falls below the 10th percentile, it suggests market recession or a lack of investor participation.

Market Regime Signals:

- **Bull Market (Regime = 1):** Short-term moving average (SMA_20) is above the long-term moving average (SMA_50).
- **Bear Market (Regime = 0):** Short-term moving average (SMA_20) is below or equal to the long-term moving average (SMA_50).

Return Expectations:

- Positive returns during high predicted volume in a bull regime signal momentum continuation (Buy Signal).
- Negative returns during high predicted volume in a bear regime signal reversal opportunities (Sell Signal).

Signal Logic: The final trading signal is determined using the following conditions:

- Buy Signal(1)
  - Volume_pred > High Volume Threshold (90th percentile)
  - Market Regime = Bull Market (1)
  - Recent Returns > 0
- Sell Signal(-1)
  - Volume_pred > High Volume Threshold (90th percentile)
  - Market Regime = Bear Market (0)
  - Recent Returns < 0
- Hold Signal(0):None of the above conditions are satisfied.

This structured method combines quantitative thresholds with market regime signals to ensure informed and precise trading decisions.

3.3.3 Position Sizing: Effective position sizing is crucial for managing risk and optimizing returns. The strategy employs dynamic position sizing based on market regimes and signal types.

Dynamic Position Sizing:

- Bull Market Buy Signal: Increase the position size to take advantage of upward momentum.
- Bear Market Sell Signal: Allocate a smaller short position to reduce exposure in unfavorable conditions.
- Hold Signal: No positions are taken, ensuring the preservation of capital during unstable periods.

Maximum Exposure Cap:

- A predefined limit prevents excessive exposure to a single trade, guarantees portfolio diversification and risk management.

This approach guarantees optimal capital allocation by aligning exposure with market dynamics and the level of confidence in signals.

3.3.4 Risk Management: Risk management mechanisms are integrated into the strategy to protect against extreme losses and ensure long-term portfolio sustainability.

Stop-Loss Mechanism:

- Positions are automatically closed if the price moves 5% against the trade direction.

Take-Profit Thresholds:

- Positions are closed once returns exceed a predefined 10% profit target.

Market Regime Adaptation:

- In bear market regimes, the strategy reduces exposure and focuses on short-selling opportunities.

These measures help maintain stability during times of high market volatility and adverse trading conditions.

3.3.5 Backtesting Framework: To validate the effectiveness of the trading strategy, a particular backtesting framework was implemented using NVDA stock data from the test dataset only.

Backtesting Platform:

- Backtests were conducted using historical NVDA stock data.

Performance Metrics:

- Sharp Ratio, Net P&L which we mentioned in 3.2.5 and finally Annualized Return which estimates long-term returns based on the test period results.

Historical Timeframe:

- The backtest period corresponds particularly to the test set, ensuring that the model operates on out-of-sample data.

3.3.6 Optimization and Tuning: **Parameter Optimization:**

- Fine-tuned volume thresholds, market regime probabilities, and LSTM model hyperparameters to maximize strategy performance.

Out-of-Sample Testing:

- Conducted tests particularly on unseen data to prevent data leakage and ensure realistic performance evaluation.

Sensitivity Analysis:

- Evaluated how strategy performance changes under different market conditions and parameter settings.

3.3.7 Implementation Plan: **Data Integration:**

- Real-time data feeds can be integrated for continuous strategy execution.

Real-Time Signal Generation:

- Model predictions and signals can be executed in a live trading environment.

Monitoring Dashboard:

- Develop dashboards to visualize key performance metrics, trading signals, and risk exposure dynamically.

3.3.8 Relevance to Study Objective: The trading strategy satisfies the volume-based trading strategy objective defined in this study.

- **Utilize Trading Volume Dynamics:** By predicting and analyzing trading volumes to inform buy and sell decisions.
- **Account for Market Regimes:** Adjusts strategy based on bull and bear market signals
- **Quantify Strategy Performance:** Evaluates outcomes using robust metrics such as Sharpe Ratio.

Expected Outcomes:

- A rule-based trading strategy tailored for NVDA stock.
- Clear, data-driven buy, sell, and hold signals.
- Robust risk management mechanisms.
- Positive performance metrics validated through backtesting.

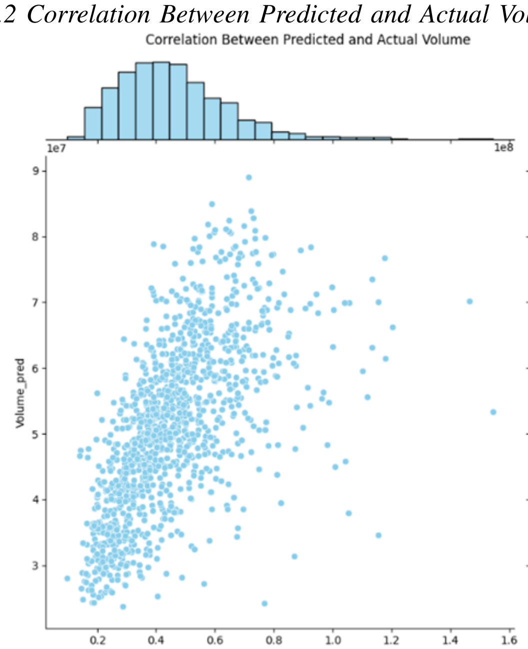
This strategy combines predictive modeling, market regime detection, and risk-adjusted trading rules to create a comprehensive volume-based trading system.

IV. Results and Discussion

4.1 Equity Curve Analysis

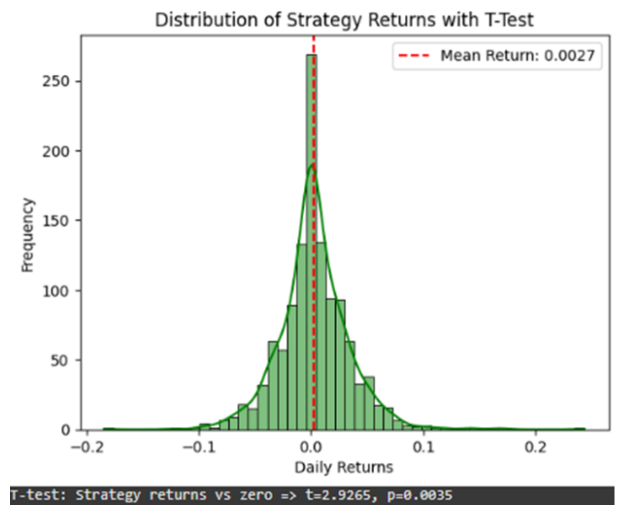
The Equity Curve demonstrates how our volume-based trading strategy has performed. It monitors the rise in capital throughout the testing period. The strategy produced signals to buy, sell, or hold, relying on volume thresholds (the 90th percentile for significant volume spikes) and different market conditions (SMA_20 compared to SMA_50) based on volumes predicted by LSTM. During the early phase, which is 2020-2021, the strategy demonstrated moderate expansion. This indicates a steady but careful approach in response to market volatility. The mid phase (2021-2022) displayed heightened volatility. It successfully controlled drawdowns using stop-loss measures and adaptable position sizing in this phase. The final phase which was held between 2023 and 2024 experienced significant capital acceleration, driven by precise volume predictions that matched optimistic market circumstances. The reason why it has achieved such a big increase in this phase is that Nvidia shares have become very popular during this period. The approach resulted in an end capital of $141, 818.50, anetprofitof$131,018.50, and a Sharpe Ratio of 1.34. These results highlight the combination of volume prediction, market condition recognition, and strong risk management, facilitating flexible and resilient trading results in diverse market environments.

4.2 Correlation Between Predicted and Actual Volume



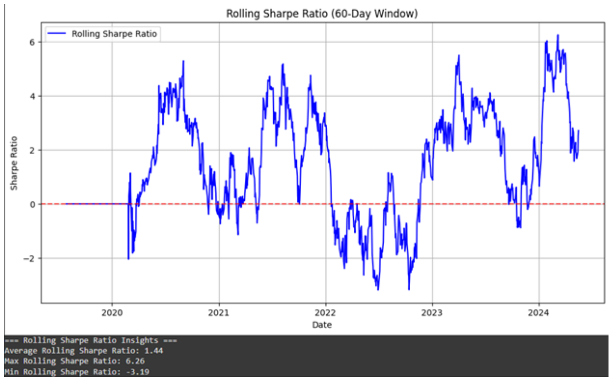
The Correlation Between Predicted and Actual Volume reflects how well our LSTM-based forecasting model captures the true patterns of trading volume. The scatter plot presented above shows a positive linear correlation between the predicted and actual volumes. It suggests that the model effectively identifies significant trends and patterns in the dataset. The calculated correlation coefficient of 0.593 indicates a moderately strong positive correlation. This confirms that the LSTM model that we created has successfully learned the dependencies among historical volume, returns, and volatility. During the early phase, the predictions were closely aligned with movements of lower volume. But in the mid phase, volatility which is increased caused slight deviations. In the final phase, as volume escalated significantly, the model succeeded in retaining prediction accuracy expect some fluctuations at extreme values. This relationship highlights the dependability of our volume predictions, which serves as the basis for future signal creation and strategy implementation. The consistency in the correlation demonstrates the strength of our feature engineering in the integration of returns and volatility as inputs for the model.

4.3 T-Test for Strategy Returns vs. Zero



The T-Test for Strategy Returns versus Zero is a statistical approach used to assess if the average daily returns produced by our volume-based trading strategy significantly differ from zero. This test evaluates whether the observed returns are a result of the trading strategy's effectiveness or are simply due to random fluctuations. In our evaluation, the average return is roughly 0.0027, as shown by the dashed red line in the histogram. The t-statistics of 2.9265 and the corresponding p-value of 0.0035 suggest statistical significance at the 0.05 confidence level, indicating that the returns differ from zero and are not merely random noise. In the early phase, the strategy displayed stable yet moderate positive returns, benefiting from the early correlation between forecasted and actual volumes. In the mid-phase, although fluctuations in market conditions and rise in volatility introduced variability in returns, the strategy consistently succeeded in yielding positive results. In the final phase, the strategy took advantage of significant market movements. This resulted in increased returns. It also contributed to the overall statistical significance seen in the T-test. This result validates the predictive power of the volume-based strategy, highlighting its ability to systematically exploit trading signals derived from predicted trading volumes, identification of market regimes, and signal generation techniques.

4.4 Rolling Sharpe Ratio Analysis



The Rolling Sharpe Ratio evaluates the strategy's risk-adjusted performance over a dynamic 60-day period, highlighting trends in performance and volatility through various market conditions. In contrast to a fixed Sharpe Ratio, this rolling method demonstrates how the strategy responds to shifting environments over time. In the early phase the ratio fluctuated around zero/ This indicates that the strategy resulted in moderate performance with occasional drawdowns. The mid-phase (2021–2023) experienced periods of exceptional returns with Sharpe Ratios near peak values, while also experiencing brief declines below zero that signaled short-term underachievement with a negative peak value of -3.19. In the final phase (2023–2024), the strategy exhibited steady positive returns with a peak value of 6.26 Sharpe Ratio, even though there were occasional drops. It ultimately stabilized at high Sharpe ratios. Typically, the strategy yielded a Rolling Sharpe Ratio of 1.44, reaching a low of -3.19, which points to times of increased risk, and a high of 6.26, demonstrating optimal performance. These findings suggest robust risk-adjusted returns overall, although certain periods may necessitate more careful observation and adjustments. The Rolling Sharpe Ratio offers essential information for fine-tuning position sizes and enhancing durability against unfavorable market fluctuations about the strategy.

V. Conclusion

This research aimed to create, execute, and verify a solid volume-based trading strategy. It aimed to use insights from both financial theory and empirical evidence. Prior studies such as Karpoff (1987), Chen (2012), and Chordia & Swaminathan (2000) have demonstrated a relation between trading volume and stock returns. They highlighted asymmetric effects across different market conditions, nonlinear dependencies, and lead-lag effects in price responses. These insights guided the development of our strategy in a better way.

Our strategy consists of LSTM-based volume forecasting, signal generation through volume and market regime thresholds, and backtesting with historical records. The testing phase concentrated directly on out-of-sample data to ensure that predictions were evaluated under realistic market conditions. Signals were generated based on three things: high and low predicted volume thresholds, the classification of regimes using moving averages, and anticipated returns. Risk management strategies, including stop-loss limits and flexible position sizing, were integrated to limit exposure. The equity curve analysis revealed strong cumulative returns and demonstrated significant growth during late testing phases because of the popularity of Nvidia dataset after 2020s. The correlation analysis between predicted and actual volumes showed a strong positive correlation (0.59). It confirms the predictive accuracy of the LSTM model. A T-Test established that daily returns were statistically significantly positive (p=0.0035), signifying consistent profitability. Lastly, rolling based sharpe ratio test showed risk-adjusted performance over a dynamic 60-day period.

In conclusion, our strategy effectively integrated machine learning, econometric methods, and financial theory to produce a rule-based trading approach with favorable performance indicators, including a Sharpe Ratio of 1.34 and a Final Capital surpassing the initial investment by 1,310%. This shows that our strategy uses Nvidia stocks as a high advantage. Although the outcomes are encouraging, future improvements, such as the implementation of dynamic thresholds or different predictive models, could further enhance performance. This research underscores the potential advantages of volume-based signals in developing adaptable, data-driven trading strategies that are capable of successfully navigating intricate financial markets.

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